

MODULE 11

Muscle Injuries & BAMIC: Sports Imaging of Muscle Pathology

BAMIC Classification · Hamstring Avulsion · Morel-Lavallée · Berlin Grade · Muscle Herniation

1.0 AMA PRA Cat 1

Sports MSK

CME Article Review

LEARNING OBJECTIVES

1. Apply the BAMIC classification to grade muscle injuries by location and severity.
2. Differentiate functional (Grade 0) from structural (Grades 1–3) muscle injuries on MRI.
3. Identify complete proximal hamstring avulsion and determine surgical vs. conservative thresholds.
4. Recognize Morel-Lavallée lesion and calcific myonecrosis on MRI and CT.
5. Apply Berlin Classification of muscle injury in the sports medicine context.
6. Identify muscle herniation, snapping hip, and other dynamic muscle pathology.

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1. BAMIC Classification — Muscle Injury Grading

The BAMIC (British Athletics Muscle Injury Classification) grades muscle injuries based on location (myofascial, muscle belly, musculotendinous junction, tendon) and severity. It is widely adopted in elite sports medicine.

BAMIC Classification — Grade and Location

Grade	Description	MRI Finding	Return to Sport
0 (Functional)	No structural damage; functional impairment only	Normal or minor physiologic edema	1–7 days (guided by symptoms)
1 (Minor)	Minor structural lesion; <10% CSA; myofascial tear	Thin hyperintense STIR signal at myofascial margin	1–2 weeks
2 (Moderate)	Partial musculotendinous tear; 10–50% CSA or >2 cm intramuscular	STIR signal in muscle belly with partial tendon disruption	3–6 weeks
3 (Severe)	Complete or subtotal tear; or avulsion	Gap sign; retraction; full CSA involved at junction	6–16 weeks or surgery
4 (Complete)	Full-thickness avulsion from bone	Gap at ischial tuberosity; bony avulsion fragment	Surgical reattachment

BAMIC Location Subtypes (letter suffix)

Suffix	Location	Example
M	Myofascial — between muscle and fascia	Strain at the fascial margin; STIR edema at periphery
Mu	Muscle belly — intramuscular	Central intramuscular tear away from the tendon
MTJ	Musculotendinous junction	Most common clinical site; proximal or distal MTJ
T	Tendon — within the tendon substance	Intratendinous signal; partial tendinopathy
P (Grade 0)	Performance — no structural lesion	Normal MRI; functional pain only

2. Proximal Hamstring Injuries — MRI Reporting

Proximal hamstring injuries range from Grade 1 myofascial strains at the conjoint tendon to complete avulsion from the ischial tuberosity. The key MRI report elements are: (1) % CSA involvement, (2) retraction distance from the ischial tuberosity, (3) sciatic nerve proximity, and (4) bone marrow edema pattern.

Proximal Hamstring Injury — MRI Assessment Checklist

Parameter	Measurement	Threshold
% CSA of conjoint tendon involved	Axial MRI at narrowest involvement	<50% = Grade 2; >50% or gap = Grade 3/4
Retraction from ischial tuberosity	Coronal or sagittal MRI	<2 cm = may manage conservatively; >2 cm = surgical
Bone marrow edema at ischial tuberosity	Coronal STIR	Present = avulsion pattern; absent = myofascial/MTJ
Sciatic nerve proximity	Axial T2 FS	Edema/contact = nerve at risk; report distance to nerve
Semimembranosus tendon status	Separate from conjoint	Semimembranosus often spared; document separately

CRITICAL

Sciatic nerve proximity is a critical surgical planning element. Scar tissue encasing the sciatic nerve post-avulsion is a major complication of delayed repair. Always assess and report the relationship of the retracted tendon to the sciatic nerve on axial images.

3. Special Entities — Morel-Lavallée, Berlin & Muscle Herniation

Morel-Lavallée Lesion — MRI Signal Characteristics

Stage	MRI Signal	Contents	Enhancement
Acute (Type 1)	T1 iso/low; T2 high	Seroma/hematoma (early)	Thin rim only
Subacute (Type 2–3)	T1 high; T2 high	Hematoma; methemoglobin	Thin peripheral enhancement
Chronic (Type 4–6)	T1 variable; T2 low-high; fibrous capsule	Fibrous capsule; hemosiderin; organized hematoma	Thick fibrous capsule; possible enhancement

Berlin Classification of Muscle Injury (2012)

Grade	Category	Structural Damage	MRI
1	Functional (Type I–II)	None — performance decrement	Normal or minor physiologic edema; no tear
2	Minor structural (Type IIIa)	Minor; <1 cm intralesional defect	Small hyperintense STIR focus; myofascial or intramuscular
3a	Moderate structural (Type IIIb)	Moderate; 1–3 cm defect	Partial tear; visible gap <50% CSA

Grade	Category	Structural Damage	MRI
3b	Severe structural (Type IV)	Complete or subtotal tear with fascial disruption	Full-thickness gap; fascial breach; retraction

TEACHING PEARL

Calcific myonecrosis: encapsulated intramuscular lesion with peripheral rim calcification, central fluid, and NO internal enhancement on CT/MRI. Can simulate sarcoma — key discriminators are chronicity, history of prior trauma or compartment syndrome, and lack of enhancement in the lesion core.

References

- Pollock N et al. BAMIC: British Athletics classification. Br J Sports Med 2014.
- Weir A et al. Muscle injuries consensus statement. Br J Sports Med 2012.
- Mellado JM, Bencardino JT. Morel-Lavallée lesion. Semin Musculoskelet Radiol 2005.